Task 1:

RAM Reseating Simulation and Documentation

Objective: Document the procedure for safely removing and re-inserting (reseating) a System RAM module, identifying potential issues observed during execution.

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1. Step-by-Step Reseating Procedure:

1. **Safety Precautions:**
   - Powered down the system completely and disconnected the main AC power cord.
   - Discharged static electricity by touching a grounded metal section of the computer case (ESD safety).
2. **Accessing RAM Slots:**
   - Removed the side panel of the PC cabinet to access the motherboard.
   - Located the DIMM (RAM) slots near the CPU socket.

3. **RAM Removal:**
    - Pushed down the retention clips/latches on both ends of the RAM slot simultaneously.
    - Carefully pulled the RAM module straight out of the slot by holding its top edges.
4. **Cleaning & Re-insertion:**
    - Inspected the slot and golden contacts for dust particles.
    - Aligned the bottom notch (key cutout) of the RAM stick with the ridge inside the motherboard slot.
    - Pressed down firmly on both ends of the RAM module until the retention clips automatically snapped into place with an audible "click".

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2. Observations & Issues Noted:

- **Improper Notch Alignment:** Inserting the RAM backward prevents it from seating and risks damaging the DIMM slot.

- **Uneven Seating Issue:** If only one side clicks, the system will fail to boot and trigger POST beep codes (e.g., 3 beeps or continuous beep) indicating memory detection failure.

Task 2:

Faulty GPU Troubleshooting Checklist

Objective: Create a structured 5-step checklist for diagnosing, troubleshooting, and verifying fixes for a faulty Graphics Processing Unit (GPU) in a gaming PC.

Troubleshooting Checklist:

- [ ] **Step 1: Symptom Identification**
    - *Actions:* Observe and document symptoms such as no display output, random black screens, visual artifacts (lines/glitches on screen), game crashes with DirectX errors, or loud GPU fan noise.
- [ ] **Step 2: Physical Connections and Power Supply Verification**
    - *Actions:* Power down the PC, turn off the PSU switch, and check all physical connections. Reseat the GPU in the PCIe slot, ensure PCIe 6+2 pin power cables are securely clicked in, and try swapping the HDMI/DisplayPort cable.

- [ ] **Step 3: Device Manager & Graphics Driver Reinstallation**
  - *Actions:* Boot into Windows (or Safe Mode) and open Device Manager. Check if the GPU displays error **Code 43** or **Code 14**. Use **Display Driver Uninstaller (DDU)** to completely wipe old drivers, then perform a clean installation of the latest NVIDIA/AMD drivers.
- [ ] **Step 4: Temperature & Stress Testing (Thermal Isolation)**
  - *Actions:* Run GPU monitoring software (e.g., GPU-Z / HWMonitor) alongside a stress test utility like **FurMark** or **3DMark**. Monitor core and hotspot temperatures to check for thermal throttling (>90°C) or thermal shutdown.
- [ ] **Step 5: Final Fix Verification**
  - *Actions:* Run a demanding game or benchmark for 30–60 minutes. Confirm stable FPS, normal operational temperatures, absence of visual artifacts, and zero system crashes to verify the fix.

Task 3:

Storage Drive Disconnection Simulation

Objective: Document BIOS status indicators and error messages observed when a system storage drive (SATA/NVMe SSD/HDD) is disconnected versus when it is properly reconnected.

1. **Observations BEFORE Reconnecting (Drive Disconnected):**

- **BIOS / UEFI Status:**
  - Under BIOS *Storage Configuration / Boot Priority*, the SATA port or M.2 NVMe slot shows **"Not Present"**, **"N/A"**, or **"None"**.
- **Startup / POST Error Messages:**
  - The system bypasses the drive and displays boot error messages on screen:
    - NO BOOT DEVICE AVAILABLE - Press any key to retry
    - Reboot and Select proper Boot device or Insert Boot Media in selected Boot device
    - Operating System Not Found

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2. Observations AFTER Reconnecting (Drive Securely Plugged In):

- **BIOS / UEFI Status:**
    - The drive model and capacity are correctly recognized in BIOS (e.g., Kingston NV2 PCIe 4.0 NVMe SSD 500GB).
    - The drive automatically reappears as the primary option under **Boot Priority Order**.
- **Startup / System Behavior:**
    - System passes POST without errors and boots straight into Windows OS within normal boot time.
    - All logical partitions (C: drive, D: drive) and saved files are fully accessible.

Task 4:

Random PC Restart Diagnostic Strategy

Objective: Select the primary component to inspect during random PC restarts, justify the choice, and detail specific diagnostic procedures.

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1. **Primary Component Selected & Reasoning:**

- **Component to Check First: System RAM (Random Access Memory)**
- **Reasoning:** Random/sudden system restarts without a Blue Screen (BSOD) or immediate thermal shutdown are most commonly caused by memory read/write corruption. When the CPU attempts to access a corrupted memory block in RAM, the operating system kernel immediately triggers an automatic reboot to prevent system-wide data corruption. While GPU and storage issues usually cause driver crashes, artifacting, or system freezes, RAM faults directly cause spontaneous system reboots.

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2. Specific Diagnostic Steps for RAM:

1. **Physical Reseating & Inspection:**
    - Shut down the PC, disconnect power, and remove all RAM modules.
    - Clean the DIMM slots with compressed air and golden contact pins with a soft eraser.
    - Reseat the RAM sticks firmly back into the motherboard slots.
2. **Run Built-in Windows Memory Diagnostic:**
    - Press Win + R, type **mdsched.exe**, and select **"Restart now and check for problems"**.
    - Allow the system to complete the basic/standard memory scan upon reboot.
3. **Perform Deep Memory Scan with MemTest86:**
    - Boot the system from a bootable USB drive containing **MemTest86**.
    - Pass at least 2 to 4 full test cycles. If any single memory error is detected, the RAM module is confirmed faulty.
4. **Isolate Faulty RAM Stick (Single-Stick Testing):**
    - If using multiple RAM sticks, test the PC using **only one RAM module at a time**.

- Run heavy applications or stress tests on each module independently to isolate which specific RAM stick causes the random restarts.